Hilbert–Huang Transform With Intelligent Noise Reduction for Passive SONAR Signal Processing

Elio Pithon Sarno Filho, Anderson Damacena Santos, Eduardo F. Simas Filho, Antonio Carlos Lopes Fernandes, Jr., José Manoel de Seixas, and Natanael N. de Moura, Jr.

***Abstract*—Ocean science plays a key role in marine exploration, encouraging the development of new methods for analyzing underwater acoustic waves. In passive SOund NAvigation and Ranging (SONAR) signal processing for military vessel detection and classification, the predominant technique is the short-time Fourier transform (STFT). However, this spectral analysis method has time–frequency (TF) resolution limitations, impacting performance in feature extraction and vessel dynamic behavior monitoring. The Hilbert–Huang transform (HHT) is an alternative to STFT, providing a data-driven TF analysis with high resolution. However, in standard HHT algorithms, estimation accuracy degrades as noise increases. This article presents a novel algorithm for HHT that computes the HHT with intelligent noise removal (HHT-INR). The proposed method is focused on passive SONAR surveillance applications, in which the information of interest usually comprises different sinusoidal components produced by the vessels' machinery and propeller system. An intelligent system based on support vector machine detects and removes noisy IMF during the EMD estimation process. Results with simulated and experimental passive SONAR signals indicate better performance than the STFT-based analysis. The HHT-INR reduces background noise and enhances resolution for analyzing vessel parameters in time-varying scenarios. The proposed method significantly improved frequency resolution in experimental signals, achieving an average reduction in spectral width of approximately 28.5 times. In addition, there was an average increase of 87.9 dB in the signal-to-noise ratio.**

***Index Terms*—Hilbert–Huang transform (HHT), intelligent classification system, passive sound navigation and ranging (SONAR), support vector machine (SVM), underwater acoustic signal (UAS) processing.**

I. Introduction

IN THE last decades, there have been significant advances in the detection and classification of acoustic sources in the underwater environment, with emphasis on the use of SOund NAvigation and Ranging (SONAR) systems. SONAR is used both in military operations, for ships and submarines identification [1], and in nonmilitary applications, for environment monitoring [2], seabed imaging [3], underwater communication [4], and in the modern fishing industry [5].

There are two types of SONAR systems: active and passive. In military applications, ships use active SONAR to detect, identify, and locate possible targets. In this case, a reference acoustic signal is emitted and its reflection is acquired. It is possible to estimate information about the underwater environment, including autonomous navigation, underwater communication, and localization [6], [7], [8], oil and gas exploration [9], and detecting damage in underwater structures [10].

Submarines in stealth operations typically employ passive SONAR for scene surveillance. Such a system is essential in defense technology for passive acoustic monitoring, especially in submarines, where invisibility and camouflage are strategic factors [11]. It also has civilian applications, including marine life monitoring [12] and underwater exploration with robots [13]. A hydrophone array often acquires the acoustic signal to allow signal-to-noise ratio (SNR) improvement and direction of arrival estimation [14].

The main challenge of passive detection is identifying and characterizing the signals of interest in a noisy underwater environment. Furthermore, acoustic noise sources may interfere with the detection and classification process. Thus, in passive SONAR, efficient signal processing techniques are essential for better characterizing the signals of interest, allowing for adequate detection and classification of targets [1].

In passive SONAR systems, signal processing algorithms traditionally rely on the short-time Fourier transform (STFT) [15], and on the wavelet transform (WT) [16]. Although widely applied, these time–frequency (TF) analysis techniques present some limitations. In the case of STFT, the sampling frequency and the time window size (number of samples) are directly associated with the expected frequency resolution, so larger windows provide better frequency resolution. However, short-time windows are required for passive SONAR to ensure stationarity. In WT's case, the mother-wavelet function's choice directly affects the effectiveness of identifying transient information [17]. These limitations may become evident when analyzing experimental passive SONAR signals, which are subject to variations in the vessels' operational conditions, different environmental noise features, and the occurrence of multiple targets in the region under surveillance.

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The Hilbert–Huang transform (HHT) is proposed in this work as an alternative TF analysis method for passive SONAR. The HHT combines empirical mode decomposition (EMD) with the Hilbert transform [18] to produce a data-driven TF representation suitable for nonstationary signals analysis. The scientific literature has reported several successful applications of HHT in problems involving nonlinear dynamical systems and nonstationary signals [19]. These applications include monitoring rotating machines based on motor vibration signals analysis [20], acoustic signal processing [21], and underwater acoustic signal (UAS) processing [22], [23].

In a previous work [24], HHT was successfully applied for passive SONAR signals analysis, presenting some advantages compared to STFT, such as high resolution in frequency estimation and better characterization of signals in time-varying vessels operation. Some limitations appear when the background noise increases, producing slight artifact oscillations in the estimated Hilbert spectrum (HS) and, in extreme cases, loss of information of interest.

This work proposes a novel algorithm for HHT estimation, which computes the Hilbert–Huang transform with intelligent noise removal (HHT-INR). The adopted strategy combines wavelet threshold denoising (WTD) and an intelligent classification system based on a support vector machine (SVM) to detect and eliminate the intrinsic mode functions (IMFs) composed mainly of noise during the EMD estimation process. The classification is carried out based on features extracted from the IMFs. Results obtained by applying the proposed method to experimental and simulated signals indicate that it is possible to extend the application of HHT to passive SONAR signals even in the presence of noise, achieving high resolution and accuracy (ACC).

The rest of this article is organized as follows. Section II addresses the proposed method. Its theoretical background is initially presented in Section II-A, comprising a brief overview of passive SONAR, HHT, and intelligent classification systems. Next, the various processing blocks that make up the proposed system are described in Section II-B. In Section III, the used experimental and simulated data sets are described along with the performance evaluation indexes. Section IV presents and discusses the obtained results. Finally, Section V brings the conclusions and perspectives for future works.

II. Proposed Method

One of the important TF analysis techniques used in passive SONAR signals for military applications is the detection of envelope modulation on noise (DEMON), which is based on STFT [25]. It estimates relevant information about the vessel propulsion system (e.g., propeller rotation speed and number of propeller blades) by accessing frequency-domain information from acoustic signals that suffer amplitude modulation during the cavitation process [26]. In this article, a new method is presented, in which the replacement of STFT by HHT in DEMON analysis provides more detailed information on stationary and nonstationary aspects of the vessels' spectral signature.

The proposed algorithm uses a novel EMD estimation procedure called HHT-INR (see Fig. 1). After the initial DEMON preprocessing, which includes bandpass filtering, AM demodulation, and subsampling, the EMD decomposition is performed. The IMFs are processed to extract features sensitive to the presence of additive noise, which are used as inputs to an SVM classifier for noisy IMFs elimination. The selected (low-noise) IMFs are filtered with Barklet WDT, and finally, Hilbert transform is applied to produce the HS and the marginal Hilbert spectrum (MHS) for TF representation.

A. Theorethical Background

1) Passive SONAR Signal Processing: A challenge in passive SONAR systems for detecting and characterizing underwater acoustic sources is that, in real conditions, the signal of interest may have low intensity and is normally contaminated by background noise.

Underwater acoustic ambient noise originates from natural and artificial (human-produced) sources, which may be underwater or at near-coast locations. Acoustic noise produced by waves and wind (known as sea-state noise) and rain stand out as the most frequent natural noise sources [27]. Sea state and rain noise are usually classified into various levels of intensity [24], [28]. Factories and boat traffic are some artificial acoustic noise sources propagating in the underwater environment. Fig. 2 shows a hypothetical scenario where a submarine monitors two acoustic sources (contacts) using a passive SONAR cylindrical hydrophone array.

An important task for passive SONAR is to classify the unknown vessels according to the emitted acoustic noise that contains information about the propeller system. Due to the vessel cavitation process, the propeller system frequency information is amplitude modulated, shifted to a high-frequency band, and often contaminated with background noise [11].

A fundamental signal processing technique in passive SONAR is DEMON [29], a TF analysis that improves SNR, allowing inferred information about the vessel propeller system, such as the number of propeller blades and the rotation speed [26]. Such information is used to classify the vessel's class and operating condition.

The standard DEMON analysis block diagram is presented in Fig. 3 [25], [30]. The input signal $s[n]$ undergoes a preprocessing stage, in which it is bandpass filtered to select the cavitation noise band [1], [24]. The signal is then demodulated to obtain information about the vessel's propulsion system in the cavitation noise envelope. After bandpass filtering, downsampling is performed as the sampling frequency F_s is much larger than the preserved frequency band [1], [24]. In the following, the STFT is applied. Then, a two-pass split-window filtering is used to reduce the background noise [31], thus obtaining the frequency spectrum $X_F(f)$ of the demodulated signal. After that, the DEMON spectrum is normalized to log scale (dB) dividing the spectrum by its maximum frequency component amplitude, and displayed.

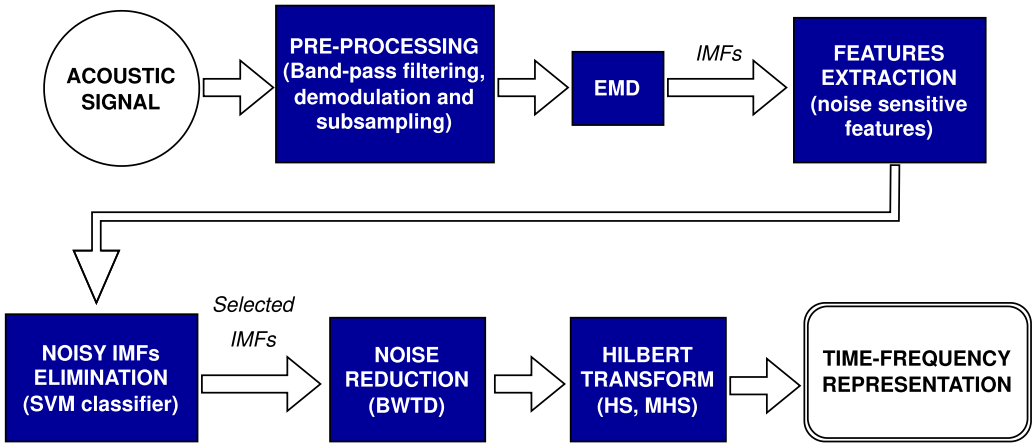


Fig. 1. Block diagram of the HHT-INR system.

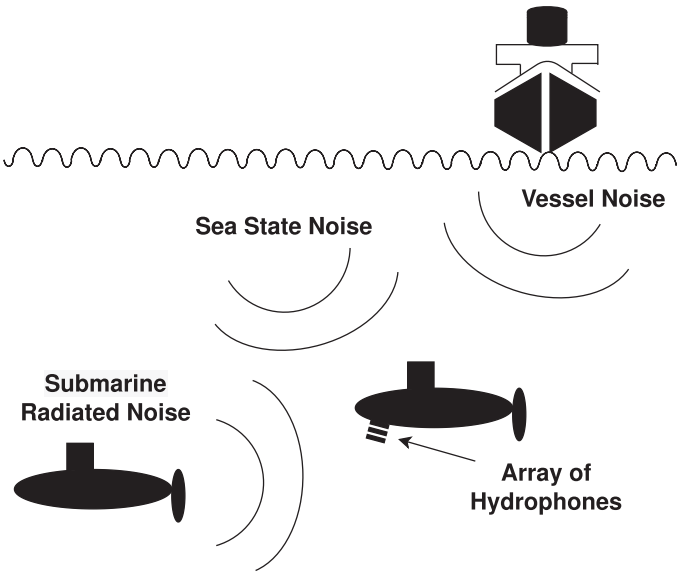


Fig. 2. Hypothetical passive SONAR system surveillance scenario. (Adapted from [11]).

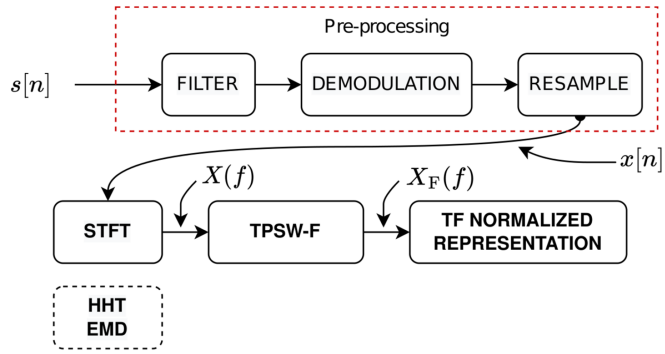


Fig. 3. Block diagram of the standard DEMON analysis, indicating a possible replacement of the STFT block by the HHT.

The result of the DEMON analysis applied to a passive SONAR experimental signal is presented in Fig. 4, as TF (spectrogram) and frequency ($\overline{P}(f)$) representations. It is possible to observe that the monitored vessel presents a shaft rotating frequency of ≈146.5 r/min, and its second harmonic (≈293 r/min) is also present with considerable intensity.

2) Hilbert–Huang Transform (HHT): HHT is an empirical TF signal analysis method with an adaptive basis [18]. Unlike Fourier analysis, in which a fixed (sinusoidal) basis is used, HHT is adaptive to local signal variations and, thus, takes into account the intrinsic behavior of the natural phenomenon [32]. As illustrated in Fig. 5, the HHT estimation comprises the following steps.

i) EMD is performed to decompose the signal of interest into a set of zero-mean orthogonal components called IMFs;
ii) Hilbert transform (HT) is applied to each IMF;
iii) HS $H(\omega, t)$ is estimated from the TF–energy information.

In EMD, the IMFs are obtained through an iterative procedure known as "sifting," in which upper and lower envelopes of the original signal $x(t)$ are defined by, respectively, interpolating the local maxima and minima using cubic splines. The mean of the upper and lower envelops (m_1) is used to define the first component (h_1)

$$h_1(t) = x(t) - m_1(t). \tag{1}$$

If $h_1(t)$ satisfies the properties of an IMF [24] (stop criterion), it is assigned as the first IMF ($c_1(t)$), and the residue is defined as $r_1(t) = x(t) - h_1(t)$. Otherwise, the procedure is repeated multiple times using as input $h_1(t)$, until the stop criterion is satisfied and the first IMF $c_1(t)$ is found.

The sifting process is then restarted using as input signal $r_1(t)$ to produce the second IMF $c_2(t)$. These iterations go on until N IMFs are found. At the end, the original signal $x(t)$ may be represented by the sum of IMFs $c_i(t)$ and one final residue $r_N(t)$ [33], as indicated in the following:

$$x(t) = \sum_{i=1}^{N} c_i(t) + r_N(t). \tag{2}$$

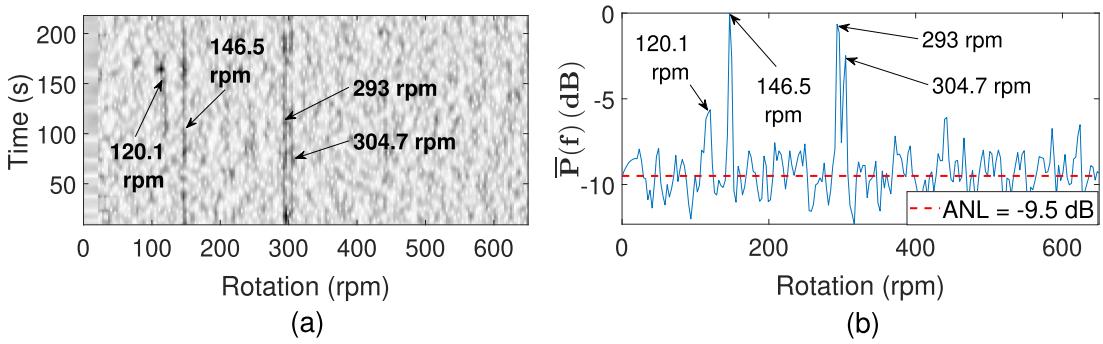


Fig. 4. Results of DEMON analysis: (a) DEMON spectrogram or DEMONGRAM. (b) Normalized average DEMON PSD, applied to a passive SONAR signal acquired from a monitoring scenario comprising at least one vessel with shaft rotating frequency of ≈ 146.5 r/min.

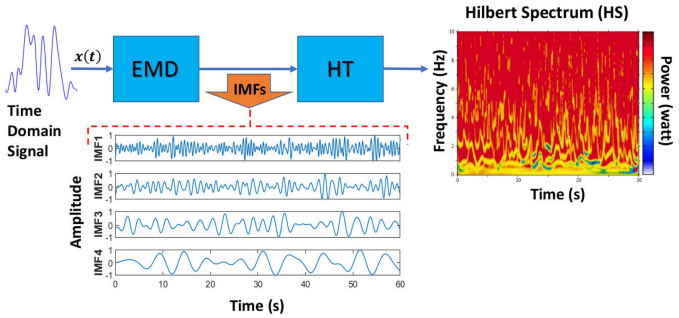


Fig. 5. HHT block diagram.

The IMFs $c_i(t)$ are used to estimate the instantaneous amplitude $a_i(t)$ [34]

$$c_i(t) + j\mathrm{HT}\{c_i(t)\} = a_i(t)e^{j\varphi_i(t)} \tag{3}$$

in which HT$\{\cdot\}$ is the HT operator

$$\mathrm{HT}\{c_i(t)\} = \int_{-\infty}^{\infty} \frac{c_i(t-\tau)}{\pi\tau} d\tau. \tag{4}$$

The instantaneous frequency $f_i(t)$ is defined as follos:

$$f_i(t) = \frac{1}{2\pi}\frac{d}{dt}\varphi_i(t). \tag{5}$$

The Hilbert amplitude spectrum (HS) $\mathrm{H}(\omega, t)$, associated with $x(t)$, is given by the superposition of individual contributions $a_i(t)$ and $f_i(t)$ of the N IMFs [33], [35], where ω is the continuous angular frequency, and $\omega_i(t) = 2\pi f_i(t)$

$$\mathrm{H}(\omega, t) = \sum_{i=1}^{N} a_i(t) \exp\left(j \int_{-\infty}^{t} \omega_i(\tau) d\tau\right). \tag{6}$$

From the Hilbert amplitude spectrum $H(\omega, t)$, it is possible to obtain the marginal HS $h_M(\omega)$ (or MHS), representing the total amplitude relative to each instantaneous frequency component $\omega(t)$, accumulated in the signal duration range $[0, T]$ [24], [33]. This is shown in the following:

$$h_M(\omega) = \int_0^T \mathrm{H}(\omega, t)\, dt, \quad \omega = 2\pi f. \tag{7}$$

EMD algorithms may present limitations, such as mode mixing (MM), which consists of oscillatory modes with very different time scales mixed in the same IMF. Considering that passive sonar signals may have relatively close harmonic components (usually within the same octave of frequency), this characteristic of EMD may become problematic, as it can prevent the separation of spectral components.

Some modifications to the original EMD algorithm have been proposed in the literature to reduce MM [24] and noise sensitivity. Among these variants, the uniform phase empirical mode decomposition (UPEMD) [36] stands out due to its greater resilience to MM caused by very close components.

In summary, UPEMD employs a set of sinusoidal component signals with n_p uniformly distributed phases in the $[0 - 2\pi]$ range and amplitude (ε_0) equal to the rms value of the added white noise. Along with the reducing MM, compared to other EMD variant [24], UPEMD showed a lower computational cost and lower residual noise [24] and mode splitting [24] values, being considered the most appropriate variant method among those analyzed for application in UAS.

In addition to the EMD variant methods, the variational mode decomposition (VMD) [24], [37] also presents interesting features, such as fast convergence and robustness to noise, having practically solved the MM problem [37]. VMD also provides good tonal separation, even for very close frequency components. Due to the above reasons, UPEMD and VMD are used in the proposed HHT-INR method.

Although suitable for analyzing nonstationary signals, EMD faces challenges when the target signal and noise share the same frequency band [38]. In specific applications, such as passive SONAR, where the relevant information is associated with sinusoidal or narrowband frequency components, and the noise is broadband, EMD may generate three types of IMFs: noisy (comprising only broadband noise), transition (or IMF-T, containing both target signal and noise), and true (narrowband signal containing significant information).

The most common ways to deal with noise in EMD estimation are 1) using a noise reduction preprocessing stage and 2) applying modified EMD versions that are more resilient to additive noise [39]. Linear filtering methods, such as Wiener filtering [40], have traditionally been used for noise reduction due to

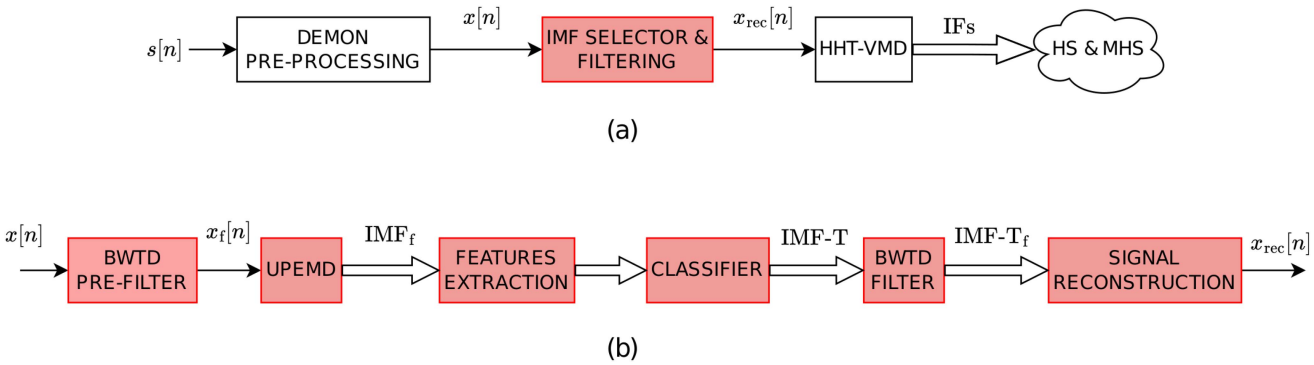


Fig. 6. (a) General block diagram of the proposed HHT-INR system. (b) Detailed IMF Selector–Filtering block diagram.

their simplicity of implementation. However, these methods are not optimized when signal and noise occupy the same frequency range. In this context, nonlinear filtering methods, such as WTD, may be applied to overcome this limitation. The Bark wavelet threshold denoising (BWTD) has been especially useful for audio and acoustic applications [41].

BWTD was used in this work for noise filtering purposes due to its better performance when compared to linear filtering methods, especially for acoustic signals where signal and noise overlap in the frequency-domain, which is the case for passive sonar. The Bark WT combines the Bark scale and wavelet denoising, providing results suitable for human auditory perception and improving the SNR in noisy conditions. Wang et al. [42] presented a structure similar to the Mel frequency cepstral coefficient, where the discrete cosine transform is replaced by Bark wavelet filters.

This work proposes a novel way to deal with noise in EMD by identifying the noisy IMFs and eliminating them from the HS computation. In this way, considering only transition and true IMFs makes the resulting TF analysis much more accurate.

3) Binary Classification Systems: A binary classification system is a model that uses input information to select between two hypotheses: H_1, normally associated with the occurrence of the event of interest, and H_0, which represents the occurrence of a nonrelevant event.

The present research investigated two classical binary classification models based on machine learning (ML) for the IMF classification problem: Fisher discriminant analysis (FDA) and SVM. The main motivation for choosing these two models was that they usually present accurate performances in scenarios where a large data set is unavailable [43], [44], as is the case here.

The FDA, also known as linear discriminant analysis, is a widely used technique for dimensionality reduction and feature extraction in ML applications and pattern classification [45]. Its basic idea consists of determining a projection direction $\mathbf{w}$ that maximizes the separation between classes.

SVM is a supervised ML technique known for its classification ACC and robustness in solving pattern recognition and regression problems [46]. Originally developed for binary classification of linearly separable patterns, SVM aims to construct a decision boundary (hyperplane) to maximize the distance between each class example. In real-world situations where patterns are not linearly separable, a nonlinear mapping of these patterns to a higher dimensional space is performed using kernel functions. The most commonly used kernel functions in SVM are polynomial, Gaussian, and sigmoidal [46].

B. Hilbert–Huang Transform With Intelligent Noise Removal (HHT-INR)

This section presents the HHT-INR for passive SONAR signals. As illustrated in Fig. 6(a), following the DEMON preprocessing of the input signal $s[n]$, the IMF Selector–Filtering block is applied to $x[n]$, obtaining a noise-reduced reconstructed version $x_{\mathrm{rec}}[n]$. Then, the HHT-VMD of $x_{\mathrm{rec}}[n]$ is performed, generating the HS and MHS representations.

We decided to use HHT-VMD as a final processing block to generate the TF representation (HS and MHS), as it offers better spectral resolution than HHT-UPEMD despite its higher noise sensitivity. This is possible as the original signal is reconstructed with reduced noise after the IMF Selector–Filtering stage.

Fig. 6(b) details the internal structure of the IMF Selector – Filtering block. Initially, a BWTD prefiltering of $x[n]$ is performed, resulting in $x_{\mathrm{f}}[n]$. Then, the UPEMD decomposition is applied to generate a set of $\mathrm{IMF_f}$. In the following, features are extracted to feed the classifier that discards the noisy IMFs, and BWTD filtering is applied to the remaining IMFs (IMF-T) for noise reduction. From the sum of the $\mathrm{IMF\text{-}T_f}$ (filtered IMF-T), the noise-reduced reconstructed version $x_{\mathrm{rec}}[n]$ of the original signal $x[n]$ is obtained. The main building blocks of the proposed system are detailed in the following topics.

1) BWTD Prefiltering and Filtering: The BWTD prefiltering block is carried out in the IMF Selector–Filtering block to increase efficiency in the feature extraction process. However, prefiltering should not be excessive to the extent that it prevents the noise characterization in the feature extraction stage. This problem is solved by properly positioning the filtering threshold in each wavelet subband.

The BWTD filtering block is applied to the IMF-T approved by the classifier, and the goal is to reduce the noise in these IMFs to obtain the $\mathrm{IMF\text{-}T_f}$.

2) Feature Extraction: It is performed in the IMF Selector–Filtering block using a time-frame structure with N_{J} windows

J_{ik} $(k = 1, 2, \ldots N_J)$ of WS samples and OL overlap factor between adjacent windows.

The five features proposed in this work are the autocorrelation function (ACF) [47], the permutation entropy (PE) [48], the spectral entropy (SE) [49], the predictivity ratio (PR) [47], and the mutual information (MI) [50], which are defined as follows.

i) The ACF measures similarity between time-series based on the lag η and is defined by (8). When normalized to 1 at $\eta = 0$, it is called the ACF coefficient $\rho_{xx}(\eta)$, defined by (9) [47]

$$r_{xx}(\eta) = \sum_{i=-\infty}^{\infty} x(i)x(i+\eta), \quad \eta = 1, 2, \ldots \tag{8}$$

$$\rho_{xx}(\eta) = \frac{r_{xx}(\eta)}{r_{xx}(0)}, \quad -1 \leq \rho_{xx}(\eta) \leq 1. \tag{9}$$

The value of $|\rho_{xx}(\eta)|$ ranges from zero to one, reaches its maximum for the zero-noise case, and tends to decrease inversely to the noise contribution [47].

ii) PE measures the randomness of a time series and allows for capturing dynamic patterns and nonlinear behaviors [51]. As described in [48], after dividing the series into fixed-size segments m, PE_{nr} is calculated using (10), where P_j is the probability of permutation of the m segments. For very noisy signals, $PE_{nr} \approx 1$, while increasingly regular signals show $PE_{nr} \approx 0$.

$$PE_{nr}(m) = \left[-\sum_{j=1}^{l} P_j \ln(P_j) \right] \Big/ \ln(m!), \quad l \leq m! \tag{10}$$

iii) The SE considers the normalized power in the frequency domain PD$[k]$ [given by (11)] as a probability distribution and computes its Shannon entropy [52]

$$PD[k] = \frac{|X[k]|^2}{\sum_{k=1}^{N} |X[k]|^2} \tag{11}$$

in which $X[k]$ is the N-point DFT of the signal $x[n]$. The normalized value of this feature (SE_{nr}) is given by [53]

$$SE_{nr} = \left[-\sum_{k=1}^{N} PD[k] \log_2(PD[k]) \right] / \log_2(N). \tag{12}$$

The values of SE_{nr} range from 0 to 1, with $SE_{nr} \approx 0$ indicating a high degree of regularity, while $SE_{nr} \approx 1$ indicates very noisy signals.

iv) The PR measures a signal's prediction ACC and also estimates its degree of randomness. The normalized value (PR_{nr}) is obtained by (13) [47], which uses a linear predicted value of the current sample: $\hat{x}[i] = \sum_{j=1}^{M} b_j\, x[i-j]$. The values of this feature range from 0 to 1, with $PR_{nr} \approx 0$ indicating low noise signals, with significant spectral components, while $PR_{nr} \approx 1$ is obtained for signals heavily corrupted by noise, making it harder to accurately predict the samples [47]

$$PR_{nr} = \sqrt{\frac{\sum_{i=i_s}^{i_e} (x[i] - \hat{x}[i])^2}{\sum_{i=i_s}^{i_e} x^2[i]}}. \tag{13}$$

v) The MI measures the statistical dependence between two random variables. In this work, we considered time-shifted windows of the signal of interest ($x(i)$ and $x(i+\eta)$) as random variables. In the discrete case, the normalized MI (MI_{nr}) is defined by (14) [54]

$$MI_{nr} = \sum_{x \in x(i+\eta)} \sum_{x \in x(i)} p(x(i), x(i+\eta)) \times \ln\left(\frac{p(x(i), x(i+\eta))}{p(x(i))p(x(i+\eta))} \right). \tag{14}$$

Ranging from 0 to 1, if $x(i)$ and $x(i+\eta)$ are independent (indicative of noisier signals), then $MI_{nr} \approx 0$ [50].

These descriptors should distinguish noisy signals from those with a higher degree of regularity, which is potentially useful for noisy IMF discrimination.

As shown in [24], EMD decomposition can be approximately modeled as a dyadic filtering process in which the useful bandwidth decreases by a factor of 2 at each decomposition level. To prevent a bias in feature estimation (as the signal's bandwidth influences the features), a successive subsampling procedure is applied at each ith EMD decomposition level. This procedure is applied only in the feature extraction block and does not affect the processing of the signal of interest.

After extracting the feature values in the N_J frames for a given signal, the median (Md) and the interquartile range (IQR) (difference between the first and third quartiles) are computed. The Md provides the central value, while the IQR measures the spread around the Md. The classifier then uses one of these two parameters (Md or IQR) as inputs.

3) IMF Classifier: The Classifier block is responsible for selecting IMF-T and eliminating the noisy IMFs. Two binary classification models, SVM and FDA, are investigated in this work. The performances of the two classifiers, using the input features arranged in groups of 2, 3, 4, or 5 and different hyperparameter values, were evaluated and compared, resulting in the choice of the best configuration, which became part of the proposed HHT-INR system.

There are several metrics for evaluating the performance of a classifier. In this work, the following were used: true positive rate (TPR, also known as probability of detection), false positive rate (FPR, or probability of false alarm), and ACC, which are defined in (15)–(17) in terms of the number of examples of the positive class (P), the number of examples of the negative class (N), the number of examples correctly classified [true positive (TP) or true negative (TN)] and incorrectly classified [false positive (FP) or false negative (FN)] [55]

$$TPR = \frac{TP}{P} \tag{15}$$

$$FPR = \frac{FP}{N} \tag{16}$$

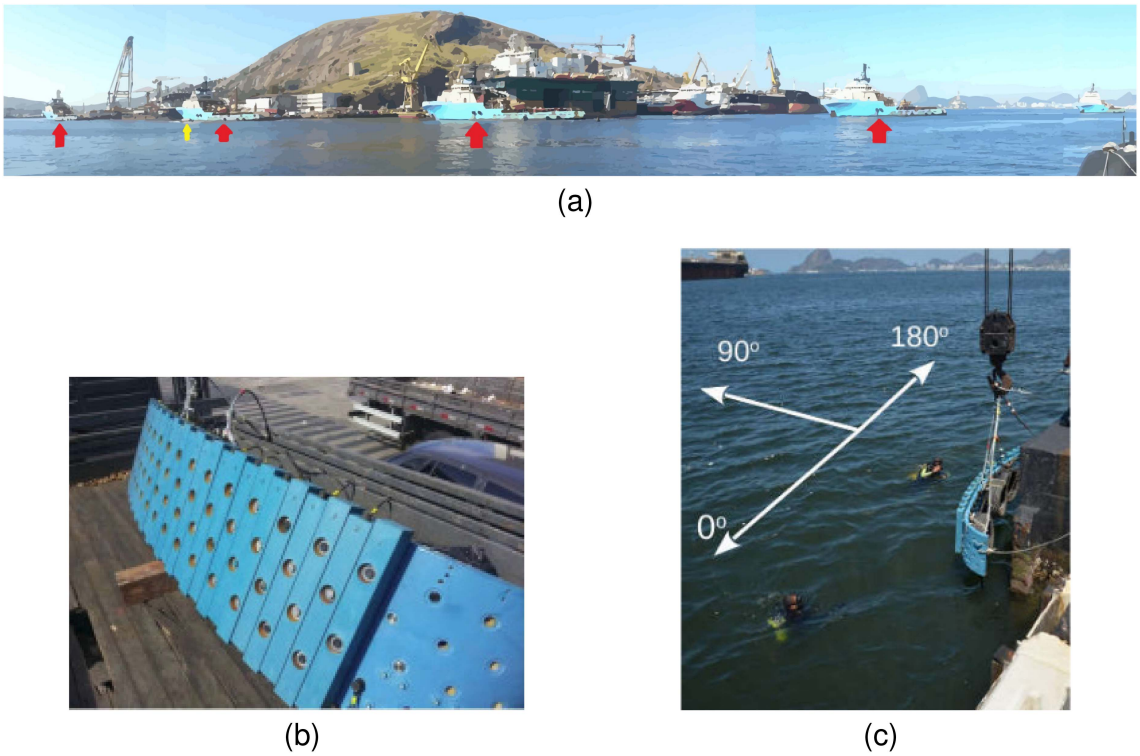


Fig. 7. Data acquisition setup used at Guanabara Bay, Rio de Janeiro, Brazil [59]. (a) Panoramic time-lapse view of Guanabara Bay, showing the passage of an offshore ship (marked with red arrows in three different positions) crossing paths with a small vessel (yellow arrow). (b) Hydrophone array. (c) Positioning of the hydrophone array concerning the ship.

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{P} + \mathrm{N}}. \tag{17}$$

The receiver operating characteristic (ROC) analysis [56] was also applied to obtain a more detailed characterization of the classifiers' behavior by varying the decision threshold and plotting the TPR $\times$ FPR curve. The area under the ROC curve (AUC) is a widely used metric and it reaches its maximum (AUC $= 1$) for the ideal case (no classification errors).

Choosing the hyperparameters for the preprocessing, feature extraction, and classification pipeline is necessary. This work adopted an exhaustive grid-search testing procedure for such a task.

III. Experimental and Simulated Data and Performance Evaluation

Experimental and simulated signals were used to validate the proposed method. The Brazilian Navy Research Institute provided two experimental signals subsets, and the third one was the ShipsEar database [57].

The simulator used in this work [58] can synthesize cavitation noise radiated by vessels and two distinct types of environmental noise (sea state and rain) at various intensity levels. Considering this, it is possible to produce signals varying the average background noise level (ANL). This allows us to evaluate the proposed method's performance under different noise conditions.

One of the Brazilian Navy data sets was acquired at Guanabara Bay, Rio de Janeiro, Brazil, using an elliptical array of 32 hydrophone channels covering a surveillance angle of 180° (see Fig. 7). A sampling frequency of $F_s = 31.250$ kHz was used [59]. The acquisition scenario consists of the passage of an offshore ship, crossing paths with a small vessel. The panoramic time-lapse image [see Fig. 7(a)] shows the ship in different positions, while Fig. 7(b) and (c) shows, respectively, the hydrophone array and its position concerning the ship's displacement [59].

The second Brazilian Navy data set was acquired in deep waters, using a cylindrical array with 24 staves for omnidirectional (360°) surveillance. The sampling frequency is the same as in the first data set. There was no information about the location of the recording. Still, it is known that a cylindrical hydrophone array was installed on a submarine, and the surveillance scenario is comprised of two other vessels. The information of interest is contaminated by self-noise (generated by the submarine itself) and sea-state noise. The results of DEMON analysis of a signal acquired in this scenario are shown in Fig. 4, comprising a vessel with shaft rotating frequency of ≈ 146.5 r/min and a harmonic rotation of ≈ 293 r/min, which characterizes a single-shaft ship with two blades. In addition, a harmonic rotation of ≈ 120.1 r/min is produced by another vessel, and the self-noise is found at ≈ 304.7 r/min.

The third set of experimental signals is part of the ShipsEar database [60], and was acquired in shallow waters at the Spanish Atlantic coast, using a linear array with three hydrophones and frequency sampling of $F_s = 52.734$ kHz [57]. The acquisition scenario is shown in Fig. 8, where the three hydrophones

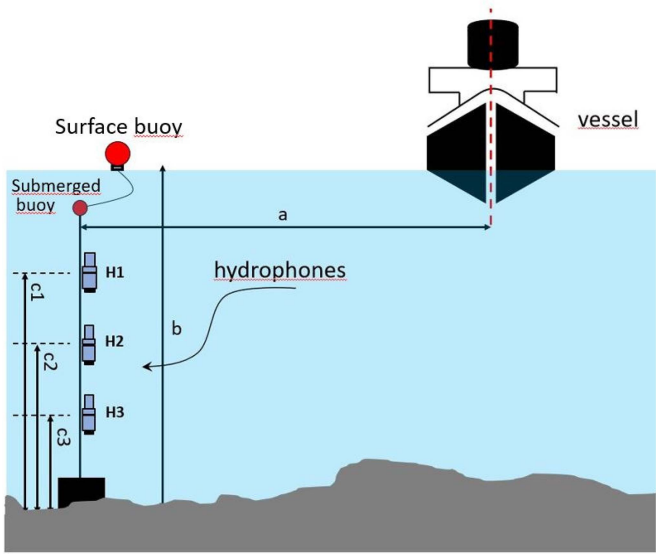


Fig. 8. Acquisition scenario of the experimental signal set from the ShipsEar database [57].

(H1–H3) can be seen positioned vertically in line at different depths (c1–c3). The hydrophones are anchored to the seabed at the Vigo port area and connected to a submerged buoy to ensure approximate verticality. A surface buoy was employed to facilitate localization and recovery. The ShipsEar data set comprises signals from several boats passing by the acquisition system in different environment noise conditions.

The final data set selection for this research comprises 40 experimental and 40 simulated signals, which emulate the main characteristics of the experimental signals, representing less rigorous noise conditions. The goal is to compare the results obtained by the proposed system when applied to real (experimental) and controlled (simulated) scenarios.

Different parameters were estimated from the passive SONAR signal TF representations for performance evaluation, as illustrated in Fig. 9. These features were chosen as they represent important features of the average spectrum, such as frequency estimation resolution and ACC, as well as the average noise level. Such scalars were computed using (18)–(20).

1) The average frequency deviation percentage $\overline{\Delta F_p}$ is used to evaluate the frequency estimation ACC in the average spectrum after DEMON preprocessing, which is given by

$$\overline{\Delta F}_{\mathrm{p}}(\text{in }\%) = \frac{1}{N_{\mathrm{h}}} \sum_{q=1}^{N_{\mathrm{h}}} \frac{\left|\widehat{F_q} - F_q\right|}{F_q} \cdot 100 \qquad (18)$$

where $\widehat{F_q}$ and F_q $(q = 1, \ldots, N_{\mathrm{h}})$ are, respectively, the estimated values of the N_{h} detected frequency components and their true values (simulation nominal values). Given the unavailability of the frequency components' true values in experimental signals, $\overline{\Delta F}_{\mathrm{p}}$ is only estimated for simulated signals.

2) The average spectral width $\overline{\Delta F_{3\mathrm{dB}}}$ estimates the frequency components resolution in the average spectrum after DEMON preprocessing, which is given by

$$\overline{\Delta F_{3\mathrm{dB}}} = \frac{1}{N_h} \sum_{q=1}^{N_h} |F_{q1} - F_{q2}| \qquad (19)$$

where F_{q1} and F_{q2} are the lateral frequencies measured 3 dB below the peak value of the component.

3) The average SNR $\overline{\mathrm{SNR}}$ (in dB) represents the mean difference between the harmonic peaks H_q and the background noise level (ANL), both in dB, in the average spectrum after DEMON preprocessing. $\overline{\mathrm{SNR}}$ may be computed by

$$\overline{\mathrm{SNR}} = \frac{1}{N_{\mathrm{h}}} \sum_{q=1}^{N_{\mathrm{h}}} \mathrm{SNR}_q = \frac{1}{N_{\mathrm{h}}} \sum_{q=1}^{N_{\mathrm{h}}} |H_q - \mathrm{ANL}| \,. \qquad (20)$$

It is important to note that to enable the comparison between the proposed method HHT-INR and the traditional DEMON analysis (based on STFT), the same frequency resolution ($\Delta f = F_{\mathrm{s}}/2048$) was used in both cases.

Different versions were compared to choose the best configuration of the signal processing chain for the proposed HHT-INR method, considering the efficiency of detecting the noisy IMFs. Human experts assigned the ground truth information for designing the IMF classifier block. Detailed results on the method design are presented in Section IV.

IV. Experimental Results

This section is organized in two parts. The first one presents the design and testing of the IMF classification block, which is one of the main building blocks of the proposed method. The second part describes the results of applying the HHT-INR method to passive SONAR signals analysis, including a comparison with the baseline DEMON algorithm.

A. Design of the IMF Classification System

The IMF classifier block is the main innovation of the proposed method. Considering this, a detailed description of the classifier design and testing procedure is provided.

The feature extraction procedure is applied to each IMF as a preprocessing step for classification. The features investigated here comprise ACF, PE, SE, PR, and MI.

The transformation from the original signal to the IMFs using EMD is heuristical, and thus, it is not guaranteed to be linear (for details see [33]). So, we are dealing with the problem of noise estimation in the heuristically transformed domain of the IMFs, which for the considered acoustic signals are composed of a combination of narrowband and broadband components. We are considering that the relevant signal information is concentrated in the narrowband tonal IMFs, as they describe the frequencies of interest for vessel characterization.

For each feature, either the Md ($\mathrm{M_d}$) or the) computed over different time windows was chosen as a descriptor based on its relevance on discriminating noisy IMFs. The choice between $\mathrm{M_d}$ and IQR was made empirically, based on the discrimination efficiency that each statistic showed for noise IMFs detection.

In the case of the prefiltering, the selection threshold Thr_m of the Bark wavelet coefficients was defined to produce a smooth

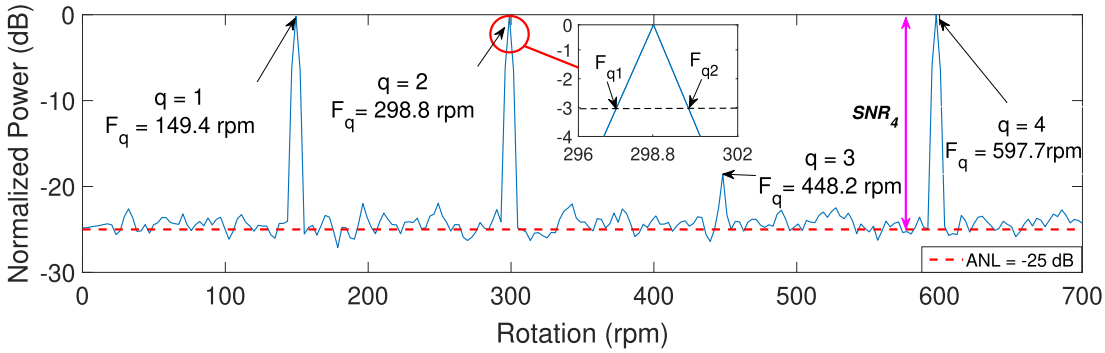


Fig. 9. Power spectrum of a generic SONAR signal with four frequency components ($N_h = 4$), indicating the graphical parameters used in the calculation of $\overline{\Delta F_{3\text{dB}}}$ and $\overline{\text{SNR}}$.

TABLE I
HYPERPARAMETER CONFIGURATIONS THAT PROVIDED THE BEST PERFORMANCE OF THE FIVE FEATURES

Atribute	WS [samples]	OL [%]	η [samples]	FPR [%]	AUC [%]
IQR^{ACF}	2350	70	250	7.62	94.4
IQR^{PE}	2048	50	-	67.62	74.3
IQR^{SE}	2048	70	-	51.43	80.3
$\text{M}_\text{d}^{\text{PR}}$	2048	70	-	76.19	59.2
$\text{M}_\text{d}^{\text{MI}}$	1024	0	-	65.71	64.2

FPR was computed for acc≈90%.

TABLE II
HYPERPARAMETERS USED IN TRAINING AND VALIDATION OF SVM CLASSIFIERS

Description	Parameters
Kernel function	linear, quadratic, cubic and Gaussian
BC	$\text{BC} = [1, 3, 5, 7]$
KS	$\text{KS} = [0.1, 0.2, 0.3, \cdots, 3.0]$

filtering, slightly reducing the background noise. For features estimation, the hyperparameters include the time window width WS, the overlap factor OL between adjacent windows, and, in the case of the ACF, the time lag used for computation [47], [61].

The feature optimization analysis varied the hyperparameters in a grid search to achieve high classification efficiency. This was done using single-variable (cut-based) classifiers by defining selection thresholds in each feature. As described in Table I, the ACF information stands out as the most discriminant metric, reaching an AUC of 94.4%. The SE feature also achieved considerable discrimination efficiency (AUC = 80.3%). These results show that for the UAS, a large amount of the information required to detect the noise in the IMFs is contained in the ACF (in other words, in the second-order statistics). The higher order statistics (accessed by MI and entropy features) seem to play a less important role in this task when looking at the individual features.

Considering the higher discrimination power of the ACF, the input feature set used for the classifier design was limited to the 15 combinations of 2, 3, 4, or 5 features, always including the IQR^{ACF}.

The grid search process for the FDA classifier, using the 15 feature combinations and the two options of the discriminant hyperparameter (linear and quadratic), revealed that the quadratic FDA with [ACF, SE] as input features outperformed the linear for all four evaluated parameters.

In the SVM classifier case, the ranges of the analyzed hyperparameters are shown in Table II, which include the kernel function, the regularization factor [box constraint (BC)], and the kernel scaling factor [Kernel scale (KS)]. After conducting a grid search and considering the same 15 combinations of features previously used, the SVM with Gaussian kernel fed from [ACF, SE] produced better results. To achieve this configuration, we used BC = 7 and KS = 0.5. The same set of features produced higher efficiencies for both classifiers.

The performance of the classifiers was compared with that of linear cuts on individual features using the ROC curves (see Fig. 10). The SVM classifier presented the best performance (larger AUC). For example, considering TPR = 90%, the FPR of SVM (2.1%) is 10.4 percent points lower than that of the FDA (12.5%). Furthermore, combining the ACF and SE features resulted in a 6.3% decrease in the SVM classifier's FPR compared to ACF individually (8.4%).

The performance parameters obtained after an iterated k-fold validation with shuffling procedure [62] are presented in Fig. 11. These results reinforce that the SVM classifier performed better than the FDA: Md ACC is 9 percent points higher, Md FPR is 1.1 percent points lower, and Md AUC is 1.9 percent points higher.

The SVM classifier performed better than the linear cuts on individual features. For example, considering the average ACC [see Fig. 11(a)], there was an increase of 4 percent points by SVM (94%) compared to ACF (90%) and 17.5 percent points compared to the value of SE (76.5%). However, concerning the average FPR [see Fig. 11(b)], both SVM and ACF linear cut presented FPR ≈ 4.2%. Finally, the SVM classifier presented an average AUC [see Fig. 11(c)] equal to 97.8%, a value nearly 1 percent point higher than ACF (96.9%) and ≈14 percent points higher than SE (83.8%).

A summary of the performance parameter values of the best-performing classifiers is presented in Table III. The SVM classifier was selected for the proposed system's IMF Selector–Filtering block. It is also worth mentioning that using a linear

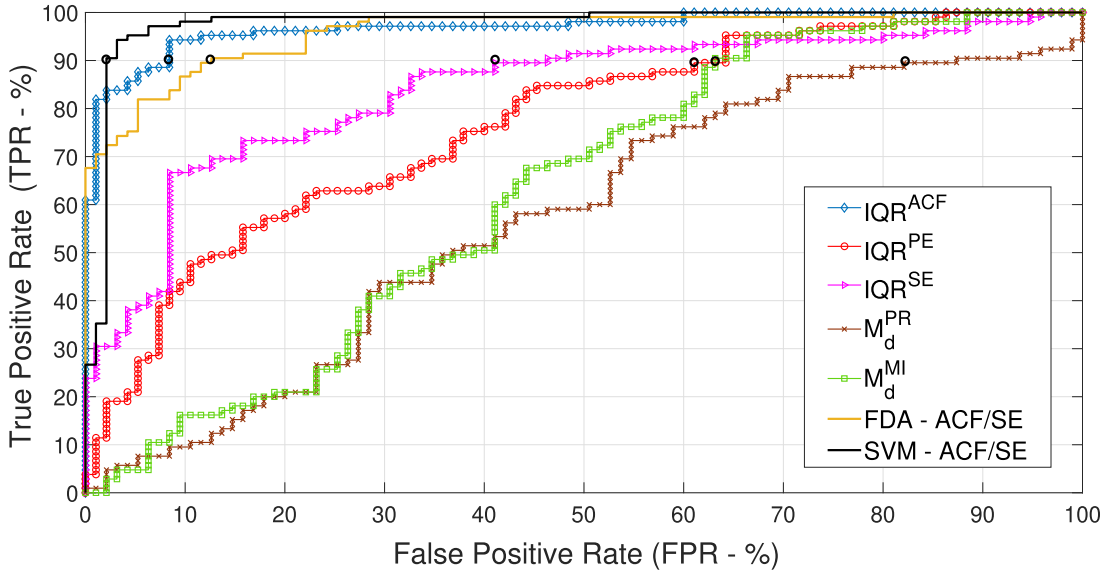


Fig. 10. ROC curves of the SVM and FDA classifiers and the five individual features. In each curve, the optimal operating point is indicated.

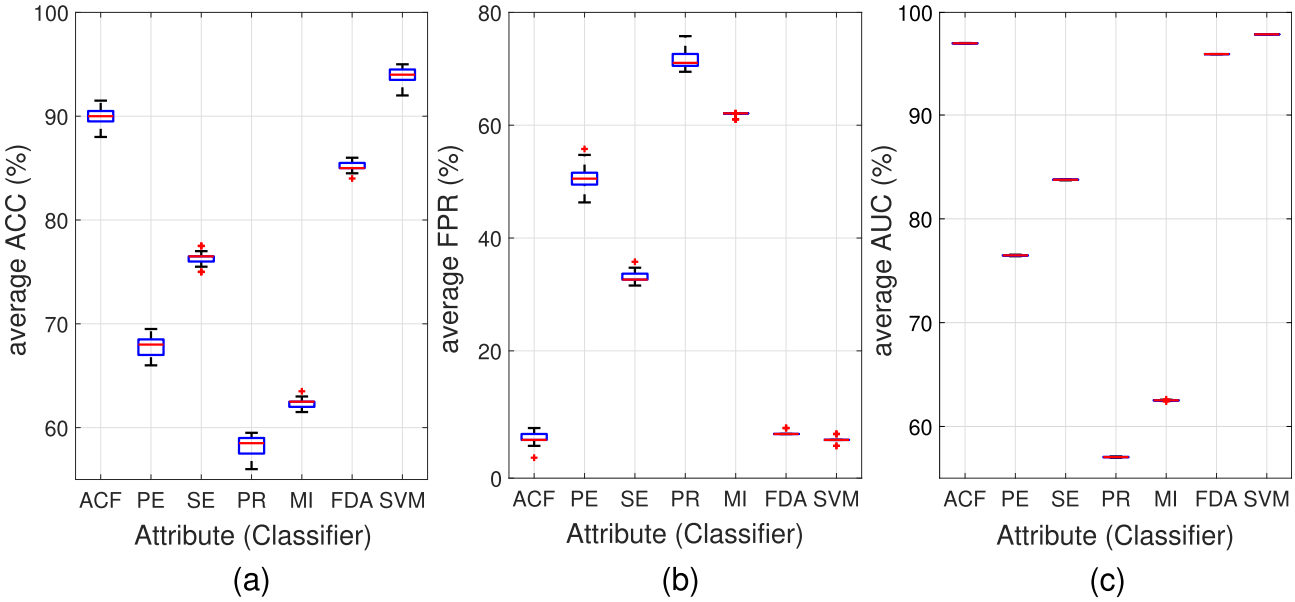


Fig. 11. Boxplots of (a) ACC, (b) FPR, and (c) AUC values for SVM and FDA classifiers and also for linear cuts applied to individual features, with indication of Md and maximum values.

TABLE III
PERFORMANCE COMPARISON OF THE BEST IMF CLASSIFIERS

Classifier	feature set	ACC (%)	FPR (%)	AUC (%)
SVM	{ACF, SE}	95.0	3.2	97.8
Linear cut	ACF	90.0	4.2	96.9

cut applied to a single feature (ACF) produces considerable efficiency for IMF classification. Such an alternative solution may be employed for special cases when the proposed system is implemented in limited computational hardware.

To better understand the ACF discrimination efficiency, Fig. 12 shows how this feature varies as a function of the ith IMF noise level (SNR_i) for the set of experimental signals. It is possible to observe that most of the IMF-T present $IQR^{ACF} > 0.04$. The classification results after SVM were also indicated in the plot, and confusion appears in some cases for $IQR^{ACF} < 0.04$, where FN and FP errors occurred. Such errors are more often associated with low SNR_i values. This feature, together with SE, was able to provide very relevant information for the SVM classifier. This reveals that higher order information (from SE) complements the second-order information (from ACF) to detect the IMF's noise contents better.

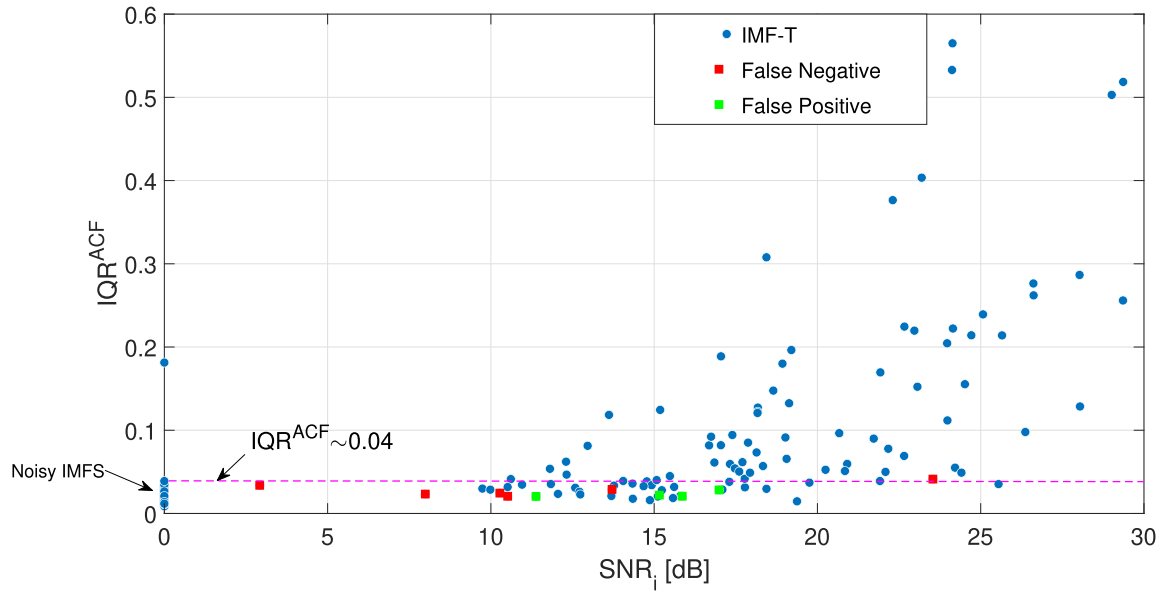


Fig. 12. IQR^{ACF} as a function of SNR_i, for the experimental data set.

TABLE IV
PERFORMANCE MEASUREMENTS OF HHT-INR, HHT-UPEMD, AND STANDARD DEMON ANALYSIS IN PROCESSING THE EXPERIMENTAL SIGNAL ACQUIRED IN GUANABARA BAY (FIG. 13), MUSSEL FISHING BOAT, AND PASSENGER SHIP (FIG. 14), BOTH FROM SHIPSEAR DATABASE

Signal	$\overline{\text{SNR}}$ (dB) ↑			$\overline{\Delta F_{3\text{dB}}}$ (r/min) ↓		
	DEMON	UPEMD	HHT-INR	DEMON	UPEMD	HHT-INR
Guanabara Bay	7.0	36.6	94.9	5.7	3.1	0.2
Mussel fishing boat	10.7	38.5	87.7	10.8	5.0	0.8
Passenger ship	5.6	42.7	63.6	14.9	3.7	0.6

The symbols ↓ (↑) Indicate that the smaller (larger) the parameter value is, the better is the method's performance.

B. Evaluation of HHT-INR for Time–Frequency Representation of Passive SONAR Signals

A performance evaluation study was conducted for the available data set (experimental and simulated). After processing 40 experimental signals (comprising a total of 200 IMFs, five per signal) using the proposed system, the performance indexes are: TPR $\approx 94.3\%$, FPR $\approx 3.2\%$, and ACC $\approx 95.0\%$ for the IMF classification block. A total of ten classification errors (5% of the IMFs) occurred, four FP, and six FN. These errors were found in specific cases, producing a very limited impact in the final TF analysis.

Fig. 13 presents a TF comparison between the traditional DEMON analysis [see Fig. 13(a)], the HHT-UPEMD algorithm [see Fig. 13(b)], and the proposed HHT-INR system [see Fig. 13(c)]. An experimental signal was used from an offshore ship crossing paths with a small vessel acquired in Guanabara Bay (Rio de Janeiro, Brazil). For this case, no IMF classification error occurred. A visual comparison indicates that HHT-INR presents better frequency resolution, narrower spectral lines in the HS, and more well-defined spectral peaks in the MHS.

The best performance of HHT-INR is reinforced in (see Table IV). In the case of the Guanabara Bay signal, the proposed system exhibited an average spectral width $\overline{\Delta F_{3\text{dB}}}$ about 28.5 times smaller and $\overline{\text{SNR}}$ with a value 87.9 dB higher compared to the DEMON analysis. Regarding HHT-UPEMD, the proposed system obtained an $\overline{\Delta F_{3\text{dB}}}$ about 15.5 times smaller and an $\overline{\text{SNR}}$ 58.3 dB higher. These parameters confirm the qualitative results from Fig. 13.

Fig. 14 brings two other experimental signals from ShipsEar database [60]. Here, the aim is to compare the TF representations provided by the traditional DEMON analysis with the proposed HHT-INR method. The first signal [see Fig. 14(a)] is produced by a mussel fishing boat featuring a four-blade propeller rotating at $\approx$ 497 r/min. The second signal [see Fig. 14(b)] is produced by a passenger ship arriving at the Port of Vigo, equipped with a two-blade propeller rotating at ≈438 r/min. In this case, there is variation in the vessel's speed, as detailed for the second harmonic component (≈876 r/min). In both signals, there is a satisfactory agreement between the frequency values of the spectral components detected by DEMON analysis (spectrogram) and those obtained by HHT-INR (HS). Similar to what occurred in the case of the Guanabara Bay signal, the results from Table IV indicate that for these two signals as well, the proposed system showed better performance compared to the DEMON analysis, exhibiting a $\overline{\Delta F_{3\text{dB}}}$ value between 13 and 25 times smaller and $\overline{\text{SNR}}$ with a value between 58 dB and 77 dB higher compared

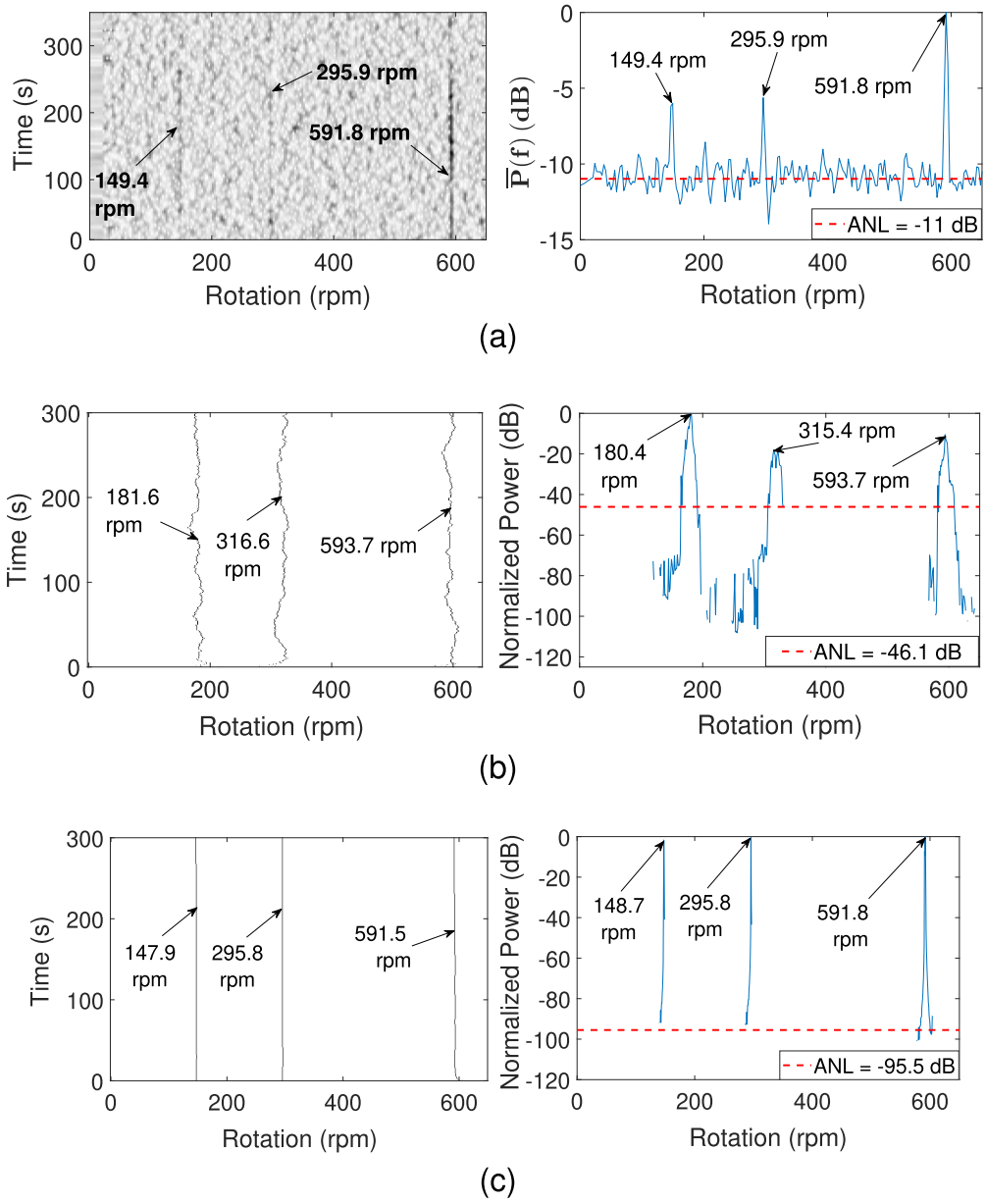


Fig. 13. (a) DEMON analysis (spectrogram on the left and $\overline{P}(f)$ on the right). (b) HHT-UPEMD (HS on the left and MHS on the right) and (c) HHT-INR (HS on the left and MHS on the right) for the experimental signal acquired in Guanabara Bay, Rio de Janeiro.

to the DEMON analysis. Regarding HHT-UPEMD, the proposed system obtained a $\overline{\Delta F_{3\text{dB}}} \approx 6$ times smaller than that obtained for the two signals and an $\overline{\text{SNR}}$ between 20 to 50 times higher.

With respect specifically to Fig. 14(b), the TF representations with expanded scale (zoom) show that the HS provides more detailed information on frequency evolution within the $[802 - 811]$ r/min range compared to the spectrogram of DEMON analysis, enabling for better insight into the speed variation dynamics of the passenger ship within this interval. The DEMON analysis, with its low-frequency resolution, could lead to a false interpretation of the spectral discontinuity at 802.3 r/min as the approach of two vessels. Conversely, the HS analysis offers higher spectral resolution, clarifying that it is the same vessel experiencing a significant speed variation of $\approx$9 r/min over $\approx$120 s. In a target tracking scenario, such time-varying information may be extremely relevant for determining the vessel displacement pattern.

In Fig. 15, the results of applying HHT-INR to the set of simulated counterparts of the 40 experimental signals are presented. Here, the simulator was adjusted to mimic the main features of the experimental signals. Using simulated signals allows us to compute the frequency estimation ACC ($\overline{\Delta F_{\text{p}}}$). There is slightly better performance of HHT-INR when compared to the traditional DEMON analysis as it presents an Md value of $\overline{\Delta F_{\text{p}}} \approx 0.23\%$, lower than that obtained with

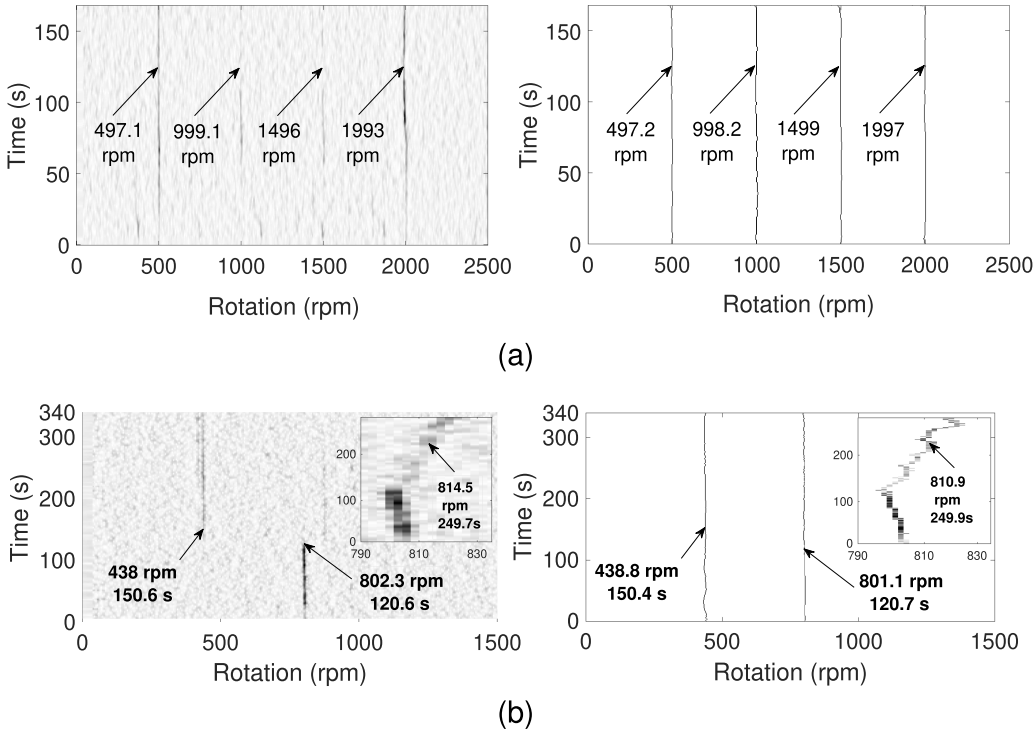


Fig. 14. Results of applying HHT-INR and standard DEMON analysis to two experimental signals from ShipsEar database [60] (DEMON analysis spectrogram on the left and HS resulting from HHT-INR on the right). (a) Mussel fishing boat. (b) Passenger ship.

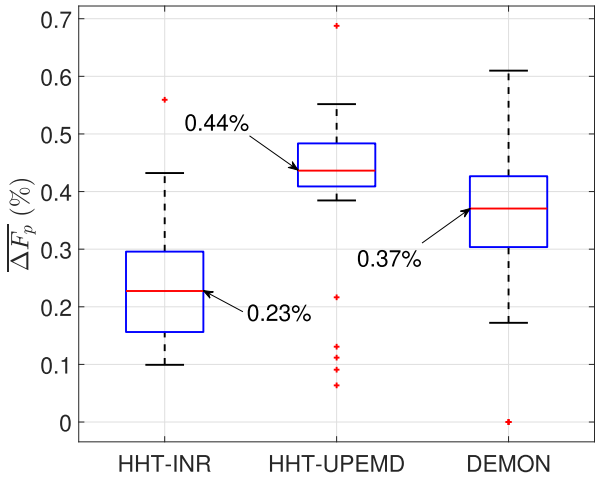


Fig. 15. Average frequency deviation ($\overline{\Delta F}_{\mathrm{p}}$) results applied to the set of 40 simulated signals. It shows the frequency estimation ACC ($\overline{\Delta F}_{\mathrm{p}}$) of HHT-INR, HHT-UPEMD, and standard DEMON analysis.

DEMON ($\overline{\Delta F}_{\mathrm{p}} \approx 0.37\%$), as well as with HHT-UPEMD ($\overline{\Delta F}_{\mathrm{p}} \approx 0.44\%$).

It is also important to analyze the few cases where classification errors occurred and their effects on the final TF representation produced by the HHT-INR method. In most cases, the classification errors do not reflect on the loss of relevant information. Still, for some special conditions, the IMF classification error does affect the final TF analysis. Considering, for example, a signal generated by a passenger speedboat with a three-blade propeller where the speed varies from ≈ 586 to ≈ 630 r/min (from ShipsEar data set), it can be seen in Fig. 16(a) its IMF decomposition. The classification results and the error type (in this case, an FN) are also pointed out. In Fig. 16(b) and (c), the normalized average PSD $\overline{P}(f)$ from DEMON and the MHS from HHT-INR are shown, respectively. Frequency components with amplitudes higher than the background noise level (ANL) are detected by HHT-INR (this is the case for tones around 630 and 1260 r/min). However, the proposed method does not see the other components present at the DEMON PSD. The IMF1 was incorrectly classified as noisy IMF and discarded, producing an FN error. Consequently, the 1888 r/min tone does not appear in the MHS. This may have occurred due to the higher level of noise incident on IMF1. The frequency component 1216 r/min, despite being detected in IMF2, is at background noise level and was eliminated by the prefilter block, similarly to the 586 r/min tone.

We evaluated the computational cost of the proposed method using the following computational setup: Windows 10 (64-bit) operating system, Intel Core i7 processor (2.9 GHz), 8 GB RAM, and GNU/Octave for implementing the routines. The execution of DEMON took in average ≈ 1.6 s while HHT-INR ≈ 8.5 s (approximately 5.3 times higher). Although the proposed system requires more time for each analysis, it is executed within a very short time interval, considering that a vessel is usually monitored in a passive SONAR system for some minutes.

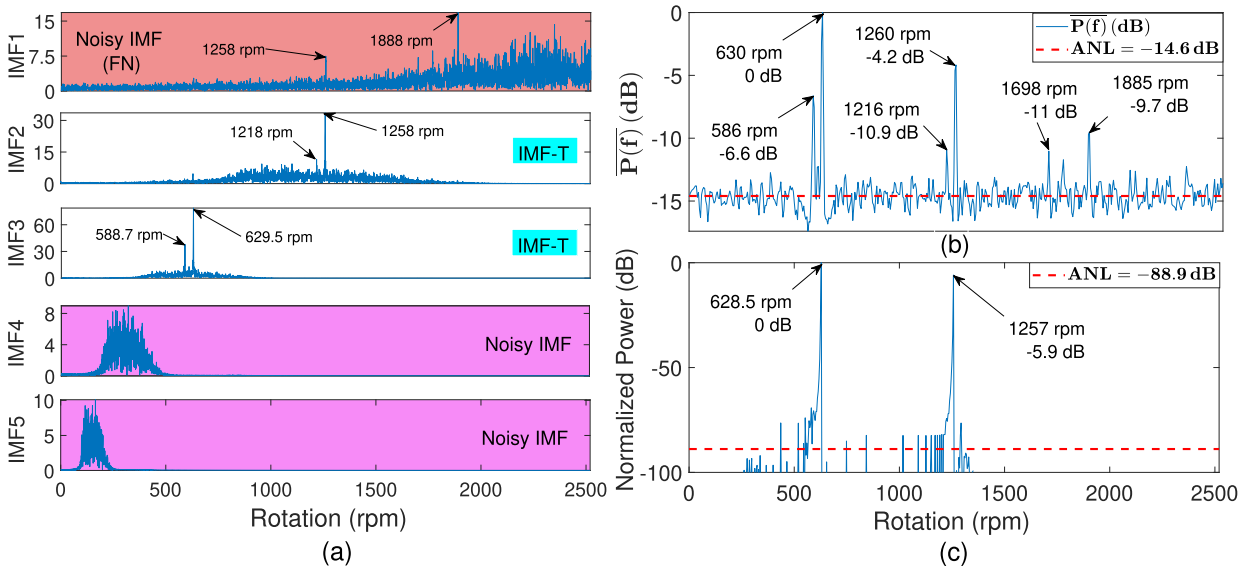


Fig. 16. Results of HHT-INR processing of the signal from the passenger speedboat. (a) Classifier outputs in the "IMF Selector-Filtering" block. (b) Normalized average PSD $\overline{P}(f)$ (DEMON analysis) and (c) MHS.

V. Conclusion

A modified HHT method for passive SONAR signal processing was presented in this article. The HHT-INR is a new method in which the IMFs resulting from the UPEMD decomposition are separated into two groups by an SVM classifier: the IMF-T, which is processed with BWTD for noise reduction, and the noisy IMFs, which are discarded. The IMFs classification is performed based on the combination of IQR^{ACF} and IQR^{SE} features and SVM classifiers achieving an ACC of 95% and FPR of 3.2% for experimental signals. The proposed method outperformed the traditional DEMON analysis, which is based on STFT.

For experimental signals occurred a considerable increase in the SNR (larger than 58 dB) and a more refined frequency resolution (at least 13.5 times smaller). Simulated signals were used to estimate the frequency estimation ACC, and also, in this analysis, the HHT-INR method showed better results, reducing an average of 1.6 times the estimation error. A possible continuation of the research comprises further studies to improve the IMF classification efficiency, including investigating deep learning methods.

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